



Nova Lake Mobile Platform

Schematic Checklist

Rev. 1.0

April 2026

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Revision History

Document Number	Revision Number	Description	Release Date
855310	0.7	Initial release	January 2026
855310	0.7.1	Revision Update only	March 2026
855310	1.0	• Updated Strap Pull-up in Table 17 on page 19	April 2026

1.0 Schematic Checklist

1.1 RCOMP Checklist

The RCOMP recommendation for various interfaces provided below:

1.1.1 RCOMP General Guidelines and Configuration

Figure 1. RCOMP Connection Diagram

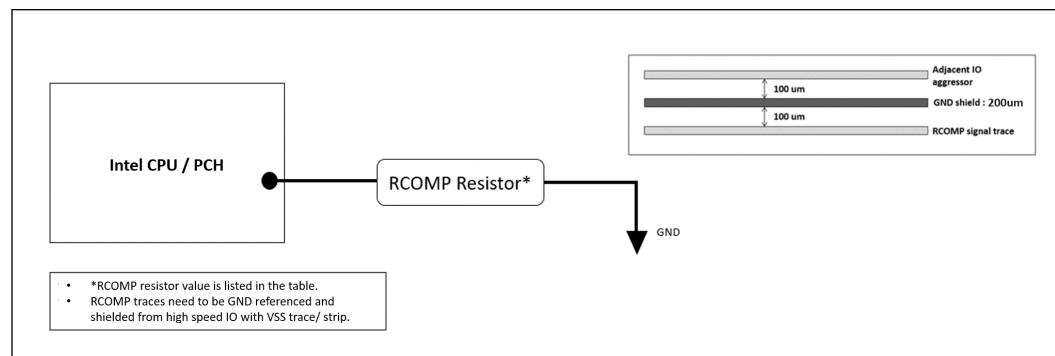


Figure 2. Example_RCOMP_with_GND_shield

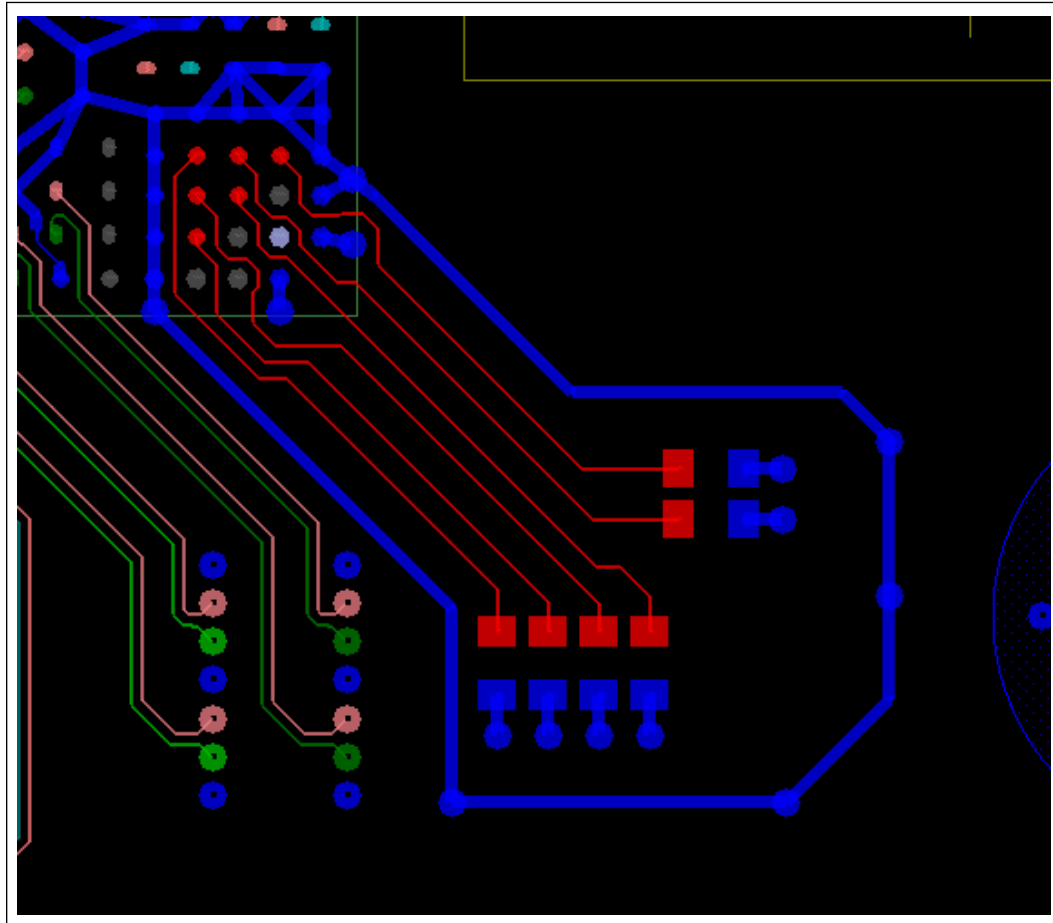


Table 1. RCOMP General Guidelines

Detail
Trace width 75um with min trace spacing 100um.
Max total length on board = 9 mm. Max number of vias 1.
GND shield to any high speed IO with VSS trace/ strip. This is not required if the adjacent signal is another RCOMP signal. PTH vias are required to be punched from this GND shield to the referenced GND plane.
GND reference required. Noisy or switching references should be avoided.
Board external resistor size=0201(recommended).
Strongly recommended to follow breakout board placement.

Table 2. RCOMP Configuration

Interface	Signal Name	Board RTerm (Ohm)	Max Board DC resistance (Ohm)	Tolerance (+/-) %	Connection on other Side	Max Board Parasitic Capacitance (pF)	Resistor Stuffing	Routing Guide
CNVi	CNV_RCOMP (CPU)	200	0.1	1	GND	1.5	Mandatory	refer to the RCOMP General Guidelines.
CSI	CSI_RCOMP (CPU)	200	0.1	1	GND	1.5	Mandatory	refer to the RCOMP General Guidelines.
Display	DDI_RCOMP (CPU)	150	0.1	1	GND	1.5	Floating	a) Unused RCOMP can be left as not connected. refer to the RCOMP General Guidelines.
PCIe	PCIe_A_RCOMP (CPU)	200	0.1	1	GND	1.5	Mandatory	refer to the RCOMP General Guidelines.
PCIe	PCIe_B_RCOMP (CPU)	200	0.1	1	GND	1.5	Mandatory	refer to the RCOMP General Guidelines.
Clock	CLK_S_RCOMP (CPU)	180	0.1	1	GND	1.5	Mandatory	Unused RCOMP can be left as not connected. Refer to the RCOMP General Guidelines.
eUSB2 and eUSB2_RCOMP	EUSB2_RCOMP (CPU)	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
USB3.2	USB32_RCOMP (CPU)	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
UFS	UFS_RCOMP	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
Soundwire	SNDW_RCOMP (CPU)	200	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.
DDR Interface	DDR_RCOMP (CPU)	100	0.1	1	GND	1.5	Mandatory	Refer to the RCOMP General Guidelines.

1.2 System Memory Checklist

1.2.1 DDR5 SODIMM/CSODIMM 1DPC Checklist

DDR5 SODIMM/CSODIMM connectivity signal should be same as RVP designs. For information, refer to the Nova Lake HX/H DDR5 SODIMM 1DPC T3 RVP Technical Documentation Kit, (# 856341).

SoC Signal Names	DDR5 SODIMM/ CSODIMM Pin Name	DDR5 SODIMM/ CSODIMM Pin Number	Notes
DDR0_CLK_[0]_N	CK0_AN	133	1
DDR0_CLK_[0]_P	CK0_AP	131	
DDR0_CLK_[1]_N	CK1_AN	134	
DDR0_CLK_[1]_P	CK1_AP	132	
DDR1_CLK_[0]_N	CK0_BN	139	
DDR1_CLK_[0]_P	CK0_BP	137	
DDR1_CLK_[1]_N	CK1_BN	140	
DDR1_CLK_[1]_P	CK1_BP	138	
continued...			

SoC Signal Names	DDR5 SODIMM/ CSODIMM Pin Name	DDR5 SODIMM/ CSODIMM Pin Number	Notes
SoC Signal Names	DDR5 SODIMM/CSODIMM Pin Name		
DDR[3:0]_CS_[1:0]	CS[1:0]_A_N, CS[1:0]_B_N		
DDR[3:0]_CA_[12:0]	CA[12:0]_A, CA[12:0]_B		
DDR[1:0]_DQ_[3:0]_[7 : 0]	DQ[31:0]_A, DQ[31:0]_B		2, 3
DDR[3:2]_DQ_[3:0]_[7 : 0]	DQ[31:0]_A, DQ[31:0]_B		
DDR[1:0]_DQSP_[3:0] DDR[1:0]_DQSN_[3:0]	DQS[3:0]_AP, DQS[3:0]_AN, DQS[3:0]_BP, DQS[3:0]_BN		4
DDR[3:2]_DQSP_[3:0] DDR[3:2]_DQSN_[3:0]	DQS[3:0]_AP, DQS[3:0]_AN, DQS[3:0]_BP, DQS[3:0]_BN		
	DM[3:0]_A_N DM[3:0]_B_N		Connecting to Ground
DRAM_RESET#			Refer to the System Memory Interface Design Guidelines Chapter in the Nova Lake H Platform Design Guide (# 855310).
DDR_RCOMP			100 Ohm resistor connected to GND
Notes: 1. CKP and CKN differential signals wapping within a pair is not allowed. Also differential clock pair to clock pair swapping within a channel is not allowed. 2. DQ Bit swapping is allowed within the same byte. 3. Byte Swapping is allowed within the same channel. 4. DQSP and DQSN differential signals wapping within a pair is not allowed. 5. Non-interleave support only. 6. Refer to the DDR5 6400 and above CKD Technical Advisory (# 845985) that provides guidance how to set the new BIOS policies for DDR5 6400 and above CLK connectivity that is a different than RVP design.			

1.2.2 LPDDR5/x Memory Down and LPCAMM2 Checklist

LPDDR5 x32 connectivity signal should be same as RVP designs. For more information, refer to the Nova Lake HX/H LP5x T4 Reference Validation Platform (RVP) Technical Documentation Kit (TDK), (# 862231).

Table 3. LPDDR5 x32 Memory Down Checklist

SoC Signal Names	LPDDR5 x32 Pin Name	Notes
DDR[3:0]_CLK_P DDR[3:0]_CLK_N	CK_C_A, CK_T_A CK_C_B, CK_T_B	
DDR[7:4]_CLK_P DDR[7:4]_CLK_N	CK_C_A, CK_T_A CK_C_B, CK_T_B	
DDR_[7:0]_CS_[1:0]	CS[1:0]_A, CS[1:0]_B	
DDR_[7:0]_CA_[6:0]	CA[6:0]_A, CA[6:0]_B	
DDR[3:0]_DQ_[1:0]_[7:0] DDR[7:4]_DQ_[1:0]_[7:0]	DQ[15:0]_A, DQ[15:0]_B	
continued...		

SoC Signal Names	LPDDR5 x32 Pin Name	Notes
DDR[3:0]_DQSP_[1:0] DDR[3:0]_DQSN_[1:0] DDR[7:4]_DQSP_[1:0] DDR[7:4]_DQSN_[1:0]	RDQS[1:0]_C_A, RDQS[1:0]_T_A, RDQS[1:0]_C_B, RDQS[1:0]_T_B	2
DDR[3:0]_WCK_[1:0]_PDDR[3:0]_W CK_[1:0]_NDDR[7:4]_WCK_[1:0]_P DDR[7:4]_WCK_[1:0]_N	WCK[1:0]_C_A, WCK[1:0]_T_A, WCK[1:0]_C_B, WCK[1:0]_T_B	
	DMI[1:0]_A and DMI[1:0]_B	Connecting to Ground
DRAM_RESET#		Refer to the System Memory Interface Design Guideline Chapter in the Nova Lake H Platform Design Guide (# 855310).
DDR_RCOMP		100 Ohm resistor connected to GND
Notes: 1. CKP and CKN differential signal swapping within a pair is not allowed. Also, differential clock pair to clock pair swapping within a channel is not allowed. 2. DQSP and DQSN differential signal swapping within a pair is not allowed. 3. LPDDR5/x supports ascend or descend option (RVP designed with descend routing signals).		

LPCAMM2 connectivity signal should be same as RVP designs. For information, refer to the Nova Lake HX LP5x LPCAMM2 T3 Reference Validation Platform (RVP) Technical Documentation Kit (TDK) (# [866574](#)).

Table 4. LPCAMM2 Checklist

SoC Signal Names	LPCAMM2 Pin Name	Notes
DDR[3:0]_CLK_P DDR[3:0]_CLK_N	LPDDR0A_CLK_DP LPDDR0A_CLK_DN LPDDR0B_CLK_DP LPDDR0B_CLK_DN LPDDR0C_CLK_DP LPDDR0C_CLK_DN LPDDR0D_CLK_DP LPDDR0D_CLK_DN	1
DDR[7:4]_CLK_P DDR[7:4]_CLK_N	LPDDR1A_CLK_DP LPDDR1A_CLK_DN LPDDR1B_CLK_DP LPDDR1B_CLK_DN LPDDR1C_CLK_DP LPDDR1C_CLK_DN LPDDR1D_CLK_DP LPDDR1D_CLK_DN	1
DDR_[7:0]CS[1:0]	LPDDR0A_CS_[1:0] LPDDR0B_CS_[1:0] LPDDR0C_CS_[1:0] LPDDR0D_CS_[1:0] LPDDR1A_CS_[1:0] LPDDR1B_CS_[1:0] LPDDR1C_CS_[1:0] LPDDR1D_CS_[1:0]	
DDR_[7:0]CA[6:0]	LPDDR0A_CA_[6:0] LPDDR0B_CA_[6:0] LPDDR0C_CA_[6:0] LPDDR0D_CA_[6:0] LPDDR1A_CA_[6:0]	
<i>continued...</i>		

SoC Signal Names	LPCAMM2 Pin Name	Notes
	LPDDR1B_CA [6:0] LPDDR1C_CA [6:0] LPDDR1D_CA [6:0]	
DDR[3:0]_DQ [1:0] [7:0] DDR[7:4]_DQ [1:0] [7:0]	LPDDR0A_DQ[1:0] [7:0] LPDDR0B_DQ[1:0] [7:0] LPDDR0C_DQ[1:0] [7:0] LPDDR0D_DQ[1:0] [7:0] LPDDR1A_DQ[1:0] [7:0] LPDDR1B_DQ[1:0] [7:0] LPDDR1C_DQ[1:0] [7:0] LPDDR1D_DQ[1:0] [7:0]	
DDR[3:0]_DQSP [1:0] DDR[3:0]_DQSN [1:0] DDR[7:4]_DQSP [1:0] DDR[7:4]_DQSN [1:0]	LPDDR0A_DQS[1:0]_DP LPDDR0A_DQS[1:0]_DN LPDDR0B_DQS[1:0]_DP LPDDR0B_DQS[1:0]_DN LPDDR0C_DQS[1:0]_DP LPDDR0C_DQS[1:0]_DN LPDDR0D_DQS[1:0]_DP LPDDR0D_DQS[1:0]_DN LPDDR1A_DQS[1:0]_DP LPDDR1A_DQS[1:0]_DN LPDDR1B_DQS[1:0]_DP LPDDR1B_DQS[1:0]_DN LPDDR1C_DQS[1:0]_DP LPDDR1C_DQS[1:0]_DN LPDDR1D_DQS[1:0]_DP LPDDR1D_DQS[1:0]_DN	
DDR[3:0]_WCK [1:0]_P DDR[3:0]_WCK [1:0]_N DDR[7:4]_WCK [1:0]_P DDR[7:4]_WCK [1:0]_N	LPDDR0A_WCK[1:0]_DP LPDDR0A_WCK[1:0]_DN LPDDR0B_WCK[1:0]_DP LPDDR0B_WCK[1:0]_DN LPDDR0C_WCK[1:0]_DP LPDDR0C_WCK[1:0]_DN LPDDR0D_WCK[1:0]_DP LPDDR0D_WCK[1:0]_DN LPDDR1A_WCK[1:0]_DP LPDDR1A_WCK[1:0]_DN LPDDR1B_WCK[1:0]_DP LPDDR1B_WCK[1:0]_DN LPDDR1C_WCK[1:0]_DP LPDDR1C_WCK[1:0]_DN LPDDR1D_WCK[1:0]_DP LPDDR1D_WCK[1:0]_DN	
	LPDDR[1:0]D_DMI[1:0]	Connecting to ground
DRAM_RESET#		Refer to the System Memory Interface Design Guideline Chapter in the Nova Lake H Platform Design Guide (# 855310).
DDR_RCOMP		100 Ohm resistor connected to GND
Notes: 1. CKP and CKN differential signal swapping within a pair is not allowed. Also, differential clock pair to clock pair swapping within a channel is not allowed. 2. DQSP and DQSN differential signal swapping within a pair is not allowed. 3. LPDDR5/x supports ascend or descend option.		

1.3 Digital Display Checklist

Table 5. eDP* /HDMI* Checklist

Function	Port	DDI Signal Names	DDI Pin Names	✓
Main Link	DDIA/DDI4	DDIx_TXP[0]	DDIx_LANE0_P	
		DDIx_TXN[0]	DDIx_LANE0_N	
		DDIx_TXP[1]	DDIx_LANE1_P	
		DDIx_TXN[1]	DDIx_LANE1_N	
		DDIx_TXP[2]	DDIx_LANE2_P	
		DDIx_TXN[2]	DDIx_LANE2_N	
		DDIx_TXP[3]	DDIx_LANE3_P	
		DDIx_TXN[3]	DDIx_LANE3_N	
AUX		DDIx_AUXP	DDIx_AUX_P	
		DDIx_AUXN	DDIx_AUX_N	
RCOMP		DDI_RCOMP		
HPD		DDSP_HPDA - DDIA DDSP_HPDx# - DDI4	HPD	
VDD Enable		VDDEN VDDEN2	VDDEN VDDEN DDI4	
BKLT Enable		BKLTEN	BKLT_EN	
BKLT Control	BKLTCTL	BRIGHTNESS		
DDC	DDPx_CTRLCLK	DDPx_CTRLCLK		
	DDPx_CTRLDATA	DDPx_CTRLDATA		
Note: x can be DDI Port A or DDI Port 4.				

1.4 Platform Clock Checklist

Table 6. Platform Clock Checklist

Signal Names	Notes	✓
CLKOUT_P/N[8:0]	Direct connect 100 MHz differential PCIe output clocks to PCIe* connector pins or device down balls.	
SRCLKREQ[8:0]#	Any used, enabled, and mapped SRCLKREQ# signal should connect to a PCIe connector pin or a device down ball with a 10 kohm $\pm 10\%$ external pull-up resistor to core rail. Any un-used, disabled, and non-mapped SRCLKREQ# signal must be left as no connects.	
XTAL_IN XTAL_OUT	See platform design guide for complete implementation details.	
CLK_S_RCOMP	See platform design guide for complete implementation details.	

1.5 CSI and Imaging Clock Checklist

Table 7. CSI Checklist (DPHY)

Signal Names	Notes
CSI_A_DP[3:0], CSI_A_DN[3:0]	Connect directly to device/connector. Polarity (P/N) cannot be reversed. In a x2 configuration, lanes [3:2] are not used
CSI_A_CLK_P, CSI_A_CLK_N	
CSI_B_DP[1:0], CSI_B_DN[1:0]	Connect directly to device/connector. Polarity (P/N) cannot be reversed. CSI_B can be used only if CSI_A is not x4
CSI_B_CLK_P, CSI_B_CLK_N	
CSI_C_DP[1:0], CSI_C_DN[1:0]	Connect directly to device/connector. Polarity (P/N) cannot be reversed.
CSI_C_CLK_P, CSI_C_CLK_N	
CSI_RCOMP	External Reference resistor compensation (200 Ω $\pm$ 1% To GND)

Table 8. Imaging Checklist (DPHY)

Signal Names	Notes
GPP_D04/ IMGCLKOUT0 / USBC_GPP_D04	Camera Clock, Connect directly to device/connector
GPP_D00/ IMGCLKOUT1 / USBC_GPP_D00	
GPP_F07/RSVD/ IMGCLKOUT2 / USBC_GPP_F07	

Table 9. CSI Checklist (CPHY)

Signal Names	Notes
CSI_A_CPHY_A0, CSI_A_CPHY_B0, CSI_A_CPHY_C0	Connect directly to device/connector. Sequence of trio (A,B,C) cannot be reversed.
CSI_A_CPHY_C1, CSI_A_CPHY_B1, CSI_A_CPHY_A1	Connect directly to device/connector. Sequence of trio (C,B,A) cannot be reversed.
CSI_B_CPHY_A0, CSI_B_CPHY_B0, CSI_B_CPHY_C0	Connect directly to device/connector. Sequence of trio (A,B,C) cannot be reversed. CSI_B can be used only if CSI_A is not T3.
CSI_B_CPHY_C1, CSI_B_CPHY_B1, CSI_B_CPHY_A1	Connect directly to device/connector. Sequence of trio (C,B,A) cannot be reversed. CSI_B can be used only if CSI_A is not T3.
CSI_C_CPHY_A0, CSI_C_CPHY_B0, CSI_C_CPHY_C0	Connect directly to device/connector. Sequence of trio (A,B,C) cannot be reversed.
CSI_C_CPHY_C1, CSI_C_CPHY_B1, CSI_C_CPHY_A1	Connect directly to device/connector. Sequence of trio (C,B,A) cannot be reversed.
CSI_RCOMP	External Reference resistor compensation (200 Ω $\pm$ 1% To GND)

1.6 USB Type-C Checklist

Table 10. Type-C High Speed Differentials Checklist

Function	Port	Signal Names	Notes
High Speed Differential	Port 0	TCP0_TX0_P	Integrated TBT: Refer to the Hayden Bridge and June Bridge Retimer reference design for Retimer circuit and Nova Lake-H Platform Design Guide (# 855310) for topology guidelines Type-C Direct(USB+DP): Refer to the Nova Lake-H Platform Design Guide (# 855310) for topology guidelines
		TCP0_TX0_N	
		TCP0_TX1_P	
		TCP0_TX1_N	
		TCP0_TXRX0_P	
		TCP0_TXRX0_N	
		TCP0_TXRX1_P	
		TCP0_TXRX1_N	
	Port 1	TCP1_TX0_P	
		TCP1_TX0_N	
		TCP1_TX1_P	
		TCP1_TX1_N	
		TCP1_TXRX0_P	
		TCP1_TXRX0_N	
		TCP1_TXRX1_P	
		TCP1_TXRX1_N	
	Port 2	TCP2_TX_P0	
		TCP2_TX_N0	
		TCP2_TX_P1	
		TCP2_TX_N1	
		TCP2_TXRX_P0	
		TCP2_TXRX_N0	
		TCP2_TXRX_P1	
		TCP2_TXRX_N1	

Table 11. Type-C Auxiliary and Sideband Checklist

Function	Port	Alternate Mode Pin Name	Thunderbolt Mode Signal Names	Notes
Auxiliary and Thunderbolt Sideband	Port 0	TCP0_AUX_N	GPP_C16 / TBT_LSX0_TXD	Refer RVP schematics for Type C Direct Auxiliary signal connect orientations. AC caps are needed for Type-C direct implementations on Auxiliary signals.
		TCP0_AUX_P	GPP_C17 / TBT_LSX0_RXD	
	Port 1	TCP1_AUX_N	GPP_C18 / TBT_LSX1_TXD	
		TCP1_AUX_P	GPP_C19 / TBT_LSX1_RXD	
continued...				

Function	Port	Alternate Mode Pin Name	Thunderbolt Mode Signal Names	Notes
	Port 2	TCP2_AUX_N	GPP_C20 / TBT_LSX2_TXD	Refer RVP schematics for LSx signal connections
		TCP2_AUX_P	GPP_C21 / TBT_LSX2_RXD	

Function	Port	Pin Name	Notes
Power	All	VBUS	VBUS, capable of up to 5A @ 36V with USB PD
		VCONN	VCONN is a power source to active cable which is capable of supplying up to 0.2A @ 5V. Unconnected CC pin will be reconfigured to VCONN when a device is plugged in.
Ground	All	GND	Ground
Configuration Channels	All	CC1	Configuration Channel 1. Used for device detection, orientation detection and Alternate Mode configuration. Unconnected pins will be reconfigured to Vconn.
		CC2	Configuration Channel 2.Used for device detection, orientation detection and Alternate Mode configuration. Unconnected pins will be reconfigured to Vconn

1.7 PCI Express* Checklist

Table 12. NVL H PCI Express* Gen4 Checklist

Signal Names	Notes	✓
PCIE_A[3:0] PCIE_B[3:0]	Connect to PERpx/PERnx PCIe* device balls through a 0402 or smaller footprint AC Coupling Capacitor in the following ranges: - Gen1 and Gen2 = 75 to 265 nF; where 100 nF is nominal value - Gen3 and Gen4= 75 to 265 nF; where 220 nF is nominal value <i>Notes:</i> - For M.2 and internal cable topologies, motherboard AC coupling capacitors are required on TX and not needed on RX since they reside on the add-in cards. - No chip pack capacitors allowed	
<i>Note:</i> Any PCI Express* Transmit and associated Receive Differential Pair not being used or implemented on the platform must be left as no connects		

Table 13. NVL H PCI Express* Gen5 Checklist

Signal Names	Notes	✓
PCIE_C[3:0] PCIE_D[7:0]	Connect to PERpx/PERnx PCIe* device balls through a 0201 AC Coupling Capacitor in the following ranges: Gen5 = 176 to 265 nF; where 220 nF is nominal value	
continued...		

Signal Names	Notes	✓
	Note: • No chip pack capacitors allowed	
Note: Any PCI Express* Transmit and associated Receive Differential Pair not being used or implemented on the platform must be left as no connects.		

1.8 USB 3.2 Guidelines Checklist

A single USB 3.2 connector will have 3.2 signals, USB 2.0 signals, and an overcurrent signal routed to it as shown in table below.

Table 14. USB 3.2 Guidelines Checklist

Signal Names	System Pull-up/ Pull-down	Notes	✓
USB32_1_TX_P USB32_1_TX_N USB32_2_TX_P USB32_2_TX_N USB32_1_RX_P USB32_1_RX_N USB32_2_RX_P USB32_2_RX_N		Tx AC cap value for USB3.2 Gen1 and Gen2: AC capacitor value 100 nF nominal (75 nF - 265 nF range). The capacitor value is an Absolute range, including part tolerances. Ensure appropriate ESD and EMI suppression devices are in place. Refer to the Universal Serial Bus USB 3.2 Design Guidelines section in Nova Lake H Platform Design Guide (# 855310) . Make sure that CMC footprint is identical to a zero Ω resistor. Unused USB ports can be left as no connect.	
USB32_RCOMP	200 Ohm +/- 1% pull-down to GND.	USB Resistor Bias, Analog connection points for an external resistor to ground. 200 Ohm +/- 1% to GND. Board Rdc: < 0.1 Ohm. Board capacitance: < 1.5 pF. GND shield to any high speed IO with VSS trace/ strip. GND reference required.	

1.8.1 USB Disabling and Termination Guidelines

If a USB port(s) is not implemented on the platform:

- USBP [x]P/N signals can be left unconnected

1.9 USB 2.0 Checklist

Table 15. USB 2.0 Checklist

Signal Names	System Pull-up/ Pull-down	Notes	✓
eUSB2P_1 eUSB2N_1 eUSB2P_2 eUSB2N_2 eUSB2P_3 eUSB2N_3 eUSB2P_4		Ensure appropriate ESD and EMI suppression devices are in place. Unused USB ports can be left as no connect due to internal pull-down resistors.	
continued...			

Signal Names	System Pull-up/ Pull-down	Notes	✓
eUSB2N_4 eUSB2P_5 eUSB2N_5 eUSB2P_6 eUSB2N_6 eUSB2P_7 eUSB2N_7 eUSB2P_8 eUSB2N_8			
GPP_E09/USB_OC0#/ USBC_GPP_E09 GPP_B11/USB_OC1#/ DDSP_HPD2/DISP_MISC3 GPP_B14/USB_OC2#/ DDSP_HPD4/DISP_MISC4 GPP_B15/USB_OC3#/USB- C_GPP_B15	10 kΩ pull-up to +1.8V based on the GPIO group.	OC0#, OC1#, OC3#: when used, 10 kΩ pull-up to 1.8V (+1.8V based on the GPIO group) is required. OC2#: when used, 100 kΩ pullup to 1.8V (+1.8V based on the GPIO group) is required. The OC2# pin is multiplexed on GPP_B14, which is a strap for Top Swap Override. A 100 kΩ pull-up ensures the strap functionality is not inadvertently asserted to enable Top Swap mode Override. Refer to the Pin Straps chapter in the EDS Vol1 for more info on the strap. When this pin is configured as GPIO, no pull-up resistor is required. Notes: • OC# pins are +1.8V tolerant . • OC# pins can be shared between USB ports. • Each USB connector should only have one OC# pin protection.	
eUSB2_RCOMP	200 Ohm +/- 1% pull-down to GND.	USB Resistor Bias, Analog connection points for an external resistor to ground. 200 Ohm +/- 1% to GND Board Rdc: <0.1 Ohm Board capacitance: < 1.5 pF GND shield to any high speed IO with VSS trace/ strip GND reference required	

1.10 CNVi Interface Checklist

Refer to the Intel® Quasar/Solar/Magnetar CNVio Schematic Checklist (# [575882](#)) for the latest CNVi schematic checklist.

Table 16. Strap Requirement Checklist

Signal Name	Strap Pull-Up/Pull-down	Notes	✓
GPP_F02/CNV_RGI_DT/ UART2_TXD	This strap has a 20 kohm $\pm$ 30% internal pull-up.	CNVi Mode Select . 0=>Integrated CNVi enabled. 1=>Integrated CNVi disabled (Default).	

1.11 Asynchronous and Sideband Signals Design Guidelines Checklist

NOTE

ES1 A0 step JTAG Debug DBX/XDP will not function without implementing onboard terminations workaround. Refer to the Nova Lake JTAG On Die Termination Workaround Technical Advisory (# [862716](#)) for details and most recent update.

Table 17. Asynchronous and Sideband Signals Design Guidelines Checklist

Signal Names	Strap Pull-Up	Notes	✓
THERMTRIP#	2.2 Kohm pull-up resistor is required connected to VCCPRIM_1P8. $55 < RLS + RODLS < 200$	Processor output signal: Asserted by the processor to indicate a thermal trip event which may cause processor severe damage, causing the processor to transition to an S5 state. If internal pull-down resistor of OD LS is 15 Ω , additional (on board) series resistor is needed to increase the total pull-down resistance of OD LS in range: $40 < RLS < 185$, in order to meet the guideline: $55 < RLS + RODLS < 200$.	
FORCEPR#	If FORCEPR used refer to the Nova Lake H Platform Design Guide (# 855310) for details on series termination resistor and Pull Up resistor and Agent PD resistor. Platform resistor value required. $R_{pu} = 2.2 \text{ k}\Omega \pm 20\%$, $75 \Omega \leq R_n + Dev_n$ drive strength $\leq 200 \Omega$. Platform resistor tolerance: 0.05 If an Agent's internal pull down resistance is $> 75\Omega$ then additional series platform resistor is not needed. If FORCEPR# is not used, then it must be terminated with a $1\text{K}\Omega \pm 5\%$ pull-up resistor to VCCPRIM_1P8.	FORCEPR# goes active when the processor temperature monitoring sensor(s) detects that the processor has reached its maximum safe operating temperature.	
PRDY#	Refer to the Chapter 4.7 in the Open and Closed Chassis Debug Nova Lake Onwards User Guide (# 870129).		
PREQ#			
CATERR#	$R_{pu} = 2.2 \text{ k}\Omega \pm 20\%$ pull-up resistor is required to be connected to VCCPRIM_1P8 .	Catastrophic Error: This signal indicates that the system has experienced a catastrophic error and cannot continue to operate.	
continued...			

Signal Names	Strap Pull-Up	Notes	✓
BPM[1:0]	Refer to the Chapter 4.7 in the Open and Closed Chassis Debug Nova Lake Onwards User Guide (# 870129).		
VIDSOUT	Rpu1 = 160 Ω, Rpu2 = 160 Ω, R1 = 10 Ω, R2 = 15 Ω connected to VCCPRIME_IO. Rpu1 and R1 are recommended to be placed close to CPU; Rpu2 and R2 are recommended to be placed close to device. Refer to the SVID Topology diagram from the Nova Lake H Platform Design Guide (# 855310) for placement and connection.	VIDALERT#, VIDSCLK, and VIDSOUT comprise a three-signal serial synchronous interface (SVID) used to transfer power management information between the processor and the voltage regulator controllers. Refer to the Nova Lake H Platform Design Guide (# 855310) implementation requirements such as location placement. Connect all pull-up resistors related to rail VDD2_CPU from CPU directly. Stuff VIDSOUT R1 with 0 Ω if VIDSOUT signal swing fails VIL with Gen5 VRTT tool. Suggest having a GND stitching vias for each signal transition. * - the value changes according to topology, refer to the Nova Lake H Platform Design Guide (# 855310) for the right value Maximum trace DC resistance 12 Ω	
VIDSCK	Rpu1 = Empty, Rpu2 = 82 Ω, R1 = 0 Ω/ Short pad, R2 = 15* Ω connected to VCCPRIME_IO Rpu1 and R1 are recommended to be placed close to CPU; Rpu2 and R2 are recommended to be placed close to device.		
VIDALERT#	Rpu1 = 100 Ω, Rpu2 = Empty, R1 = 0 Ω/ Short pad, R2 = 0* Ω/ Short pad. Rpu1 and R1 are recommended to be placed close to CPU; Rpu2 and R2 are recommended to be placed close to device. Refer to the SVID Topology diagram from the Nova Lake H Platform Design Guide (# 855310) for placement and connection. Need Rpu1=100 Ohm connected to VCCPRIME_IO.		
DBG_PMODE	Refer to the Chapter 4.7 in the Open and Closed Chassis Debug Nova Lake Onwards User Guide (# 870129).		
SOC_JTAG_TRST#			
SOC_JTAG_TMS			
SOC_JTAG_TDO			
SOC_JTAG_TDI			
SOC_JTAG_TCK			
PCD_PCH_JTAGX			
BOOTHALT#			
EPD_ON_IN	None	EPD_ON_OUT needs to be connected through 0 Ohm resistor to EPD_ON_IN	
EPD_ON_OUT			

1.12 Mother Board Acoustic Noise Consideration Checklist

- **Capacitor Size:** Increase the use 0402 capacitors for VCCPCORE, VCCECORE, VCCSA, and VCCGT instead of larger-sized capacitors.

- **Symmetric Placement:** For larger-sized capacitors, ensure the implementation is placed symmetrically. Utilize through-hole (TH) vias to reduce offset. While symmetric placement is recommended for 0402 capacitors, it is not mandatory.
- **VR_VIN Caps:** Symmetric placement is recommended for VR_VIN capacitors.
- **Additional Recommendations:** Further recommendations can be found in the Nova Lake H Platform Design Guide (# 855310) under the Acoustic Noise Mitigation Options and in the Acoustic Noise Mitigation Guidance Technical Advisory (# 575216).

1.13 RTC Circuit Checklist

Table 18. RTC Checklist

Signal Names	Notes	✓
RTCX1, RTCX2	See platform design guide for complete implementation details.	
RTCST#	An RC delay circuit with a time delay in the range of 18-25 ms should be provided. The recommended values for resistor and capacitor are 20 kohm and 1.0 μ F. The circuit should be connected to VCC RTC.	
SRTCST#	An RC delay circuit with a time delay in the range of 18-25 ms should be provided. CRB uses 20 kohm and 1.0 μ F for resistor and capacitor. The circuit should be connected to VCC RTC	
INTRUDER#	A 1 Mohm pull-up is used on the customer reference board (CRB). This is needed to reduce leakage from Coin Cell Battery in G3 state.	
Note: For further information on the RTC, refer to the Real Time Clock Section in the Nova Lake H Platform Design Guide (# 855310).		

1.14 eSPI Checklist

Table 19. Strap Requirement Checklist

Signal Names	Strap Pull-Up/Pull-down	Notes	✓
GPP_H00	This strap has a 20 kohm $\pm$ 30% internal pull-down.	eSPI Flash Sharing Mode. 0=>Host Attached Flash Sharing (aka MAFS) enabled (Default). 1=>Device Attached Flash Sharing (aka SAFS) enabled.	
GPP_C05/SML0ALERT#	This strap has a 20 kohm $\pm$ 30% internal pull-down.	eSPI Disable . 0 = Enable eSPI (Default). 1 = Disable eSPI.	

Table 20. eSPI Checklist

Signal Names	System Implementation	Notes
ESPI_IO[3:0]	16/12/22 Ohm series resistor	1-Load Topology [1] R1 : 16 Ω $\pm$ 5%. For max total length > 152.4mm [2] R1 : 22 Ω $\pm$ 5%. For max total length $\leq$ 152.4mm To be placed 25.4 mm from the processor on ESPI_IO[3:0].
ESPI_CLK	16/12/22 Ohm series resistor.	1-Load Topology [1] R1 : 12 Ω $\pm$ 5%. For max total length > 152.4mm [2] R1 : 22 Ω $\pm$ 5%. For max total length $\leq$ 152.4mm To be placed 25.4 mm from the processor on ESPI_CLK.
ESPI_RESET# ESPI_CS0#	No series resistor needed.	

1.15 Audio Checklist

Table 21. Intel® HD Audio Checklist

Signal Names	System Pull-up / Pull-down	Series Termination Resistor	Notes	✓
HDA_RST#	None	0 Ohm	Connected between Processor and Intel® High Definition Audio (Intel® HD Audio) codec. A series-termination-resistor is required for primary codec.	
HDA_SYNC	Internal pull- down	0 Ohm	Connected between Processor and Intel® HD Audio codec. A series-termination-resistor is required for the primary codec.	
HDA_BCLK	None	0 Ohm	Connected between Processor and Intel® HD Audio codec. HDA_BCLK (24 MHz) is output from the Processor to any codec present on the link.	
HDA_SDO	None	0 Ohm	Connected between Processor and Intel® High Definition Audio (Intel® HD Audio) codec. A series-termination-resistor is required for primary codec.	
HDA_SDI[1:0]	Internal Pull- Down	33 Ohm	Connected between Processor and HD Audio codec. Can be left NO CONNECT if not used	
continued...				

Signal Names	System Pull-up / Pull-down	Series Termination Resistor	Notes	✓
			(internal pull-down resistors that are always enabled). There is a weak integrated pull-down resistor on the HDA_SDI[1:0] pins.	

Table 22. Intel® DMIC Interface Checklist

Signal Names	System Pull-up / Pull-down	Series Termination Resistor	Notes	✓
DMIC_CLK_A[1:0]	None	Series resistor value changes based on the topology and additional details, refer to the Nova Lake H Platform Design Guide (# 855310).	Connect each clock signal to up to two DMIC devices (stereo mics for each DATA signal) with series resistor connected close to Processor. An additional 27 pF cap placeholder recommended to be placed as close to series resistor in case EMI/RFI protection is needed.	
DMIC_DATA[1:0]	Internal pull- down	Series resistor value changes based on the topology and additional details, refer to the Nova Lake H Platform Design Guide (# 855310).	Connect each data signal to up to two DMIC devices (stereo mics for each DATA signal) with series resistor connected close to DMIC device.	

Table 23. Intel® I²S Interface Checklist

Signal Names	System Pull-up / Pull-down	Series Termination Resistor	Notes	✓
I2S[2:0]_SFRM	Internal Pull- Down	Series resistor value changes based on the topology and additional details, refer to the Nova Lake H Platform Design Guide (# 855310).	Connected between Processor and I ² S devices. Series resistor should be connected close to the Processor/ Connector or Device.	
I2S[2:0]_SCLK	Internal pull- down			
I2S[2:0]_TXD	None			
I2S[2:0]_RXD	Internal pull- down			
I2S_MCLK1_OUT	None			

Table 24. Soundwire * Interface Checklist

Signal Names	System Pull-up / Pull-down	Series Termination Resistor	Notes	✓
SNDW[3:0]_CLK	None	Series resistor value changes based on the topology and additional details, refer to the Nova Lake H Platform Design Guide (# 855310).	Connected between Processor and SoundWire devices. Series resistor should be connected close to the Processor.	
SNDW0_DATA0	Internal pull-down			
SNDW1_DATA0				
SNDW2_DATA0				
SNDW2_DATA1				
SNDW2_DATA2				
SNDW2A_DATA1				
SNDW3_DATA0				
continued...				

Signal Names	System Pull-up / Pull-down	Series Termination Resistor	Notes	✓
SNDW3_DATA1 SNDW3_DATA2 SNDW2_DATA3				

1.16 I2C/I3C Checklist

Table 25. I2C/I3C Checklist

Signal Names	System Pull-up/ Pull-down	Notes	✓
I2C[5:0]_SCL, I2C[5:0]_SDA, I2C[3:5]A_SCL, I2C[3:5]A_SDA ISH_I2C[2:0]_SCL, ISH_I2C[2:0]_SDA, ISH_I2C2A_SCL, ISH_I2C2A_SDA THC_I2C[1:0]_SCL, THC_I2C[1:0]_SDA I3C[3:0]_SCL, I3C[3:0]_SDA, I3C[1:0]A_SCL, I3C[1:0]_SDA, I3C3A_SCL, I3C3A_SDA, I3C1B_SCL, I3C1B_SDA ISH_I3C[1:0]_SCL, ISH_I3C[1:0]_SDA	May require Internal Pull-down resistor and current assist along with External pull-up resistor.	The external pull-up resistor and current assist value depends on the speed supported and total bus capacitance. For more details, refer to the I2C / I3C chapter in the Nova Lake H Platform Design Guide (# 855310).	

1.17 SMLink Interface and System Management Checklist

Signal Names	System Pull-up/ Pull-down	Notes	✓
SML0ALERT#	External Pull-up resistor required		
SML1ALERT#	External Pull-up resistor required		
SML0BALERT#	External Pull-up resistor required		
PSE_SMLALERT#	External Pull-up resistor required		
SML[0:1]DATA, SML0BDATA, SML[0:1]CLK, SML0BCLK, USB-C_SMLCLK, USB-C_SMLDATA, PSE_SMLCLK, PSE_SMLDATA	Multiple options for the pull-up /pull- down resistor and current assist strengths are provided for a particular Total bus capacitance. Choose an optimal value for the system based on the design considerations like power dissipation, BoM cost. Additional circuitry	External Pull-up, internal Pull-down value and current assist is determined by bus characteristics. USB-C_SMLCLK, USB-C_SMLDATA is to be used for PD to PMC communication	
continued...			

Signal Names	System Pull-up/ Pull-down	Notes	✓
	may be required to connect high and low powered sections.		
PMCALERT#	Refer to section 1.29 Power Management checklist for Pull-up / Pull-down requirements. Check this regardless of whether Integrated USB- C is used or not.	Connect Alert signal from the PD chipset to the PMCALERT# Pin If Integrated USB Type C is used.	

1.18 Serial Peripheral Interface (SPI) Checklist

Table 26. Strap Requirement Checklist

Signal Names	Strap Pull-Up/Pull-down	Notes	✓
GPP_V17	This strap has a 20 kohm $\pm$ 30% internal pull-down.	Flash Descriptor Security Override. 0 - Enable security measures defined in the Flash Descriptor (Default). 1 - Disable Flash Descriptor Security (override). This strap should only be asserted high using external Pull-up in manufacturing/debug environments ONLY. The internal pull-down is disabled ~2 RTC clocks after the strap sampling event.	
GPP_H01	This strap has a 20 kohm $\pm$ 30% internal pull-down.	Enable/Disable SPI Flash Descriptor Recovery . Flash Descriptor Recovery for NIST SP800-193. 0 - Flash descriptor recovery disable (Default). 1 - Flash descriptor recovery enable The internal pull-down is disabled ~2 RTC clocks after the strap sampling event.	
SPI0_IO2	This strap has a 20 kohm $\pm$ 30% internal pull-up.	This strap should sample HIGH. There should NOT be any on-board device driving it to opposite direction during strap sampling.	
SPI0_IO3	This strap has a 20 kohm $\pm$ 30% internal pull-up.		

Signal Names	System Pull-up/Pull- down	Series Termination Resistor	Notes
SPI0_MISO, SPI0_MOSI, SPI0_CLK, SPI0_IO2, SPI0_IO3		4 / 5 / 18 / 22 / 30 / 51 Ohm	SPI input/output data line between the processor and SPI flash device. SPI0 1-Load MAF (Device Down) Topology: [1] R1 (50 MHz) - 18 Ω $\pm$ 5%. To be placed within 30.48 mm from the device on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R1 (80 MHz) - 18 Ω $\pm$ 10%.

Signal Names	System Pull-up/Pull-down	Series Termination Resistor	Notes
			To be placed within 12.7 mm from the device on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 2-Load Branch MAF (Device Down) with Flash Only Topology: [1] R1 - DATA: $30\ \Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: $22\ \Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_CLK. [2] 5 Ω. To be placed on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. It is optional to have R2 on the channel. It can be removed to reduce BOM cost. SPI0 2-Load Branch MAF (Device Down) with Flash and TPM Topology: [1] R1 (50 MHz) $30\ \Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R2 (50 MHz) $51\ \Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [3] R3 (50 MHz) 5 Ω. To be placed within 7.62 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. It is optional to have R2 on the channel. It can be removed to reduce BOM cost. [4] R1 (80 MHz) $10\ \Omega \pm 10\%$. To be placed within 12.7 mm from the device on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [5] R2 (80 MHz) $330\ \Omega \pm 10\%$. To be placed within 2.54 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [6] R3 (80 MHz) No R3 required (0 Ω/short pad). SPI0 3-Load Branch MAF (Device Down) Topology: [1] R1 - $30\ \Omega \pm 5\%$. To be placed within 10.16 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R2 - $51\ \Omega \pm 5\%$. To be placed on within 10.16 mm from the branch SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [3] R3 - 4 Ω.

Signal Names	System Pull-up/Pull-down	Series Termination Resistor	Notes
			To be placed within 10.16 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. It is optional to have R3 on the channel. It can be removed to reduce BOM cost. SPI0 80MHz 3-Load Branch MAF (Device Down) Topology [1] R1 - 10 $\Omega \pm 10\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R2 - 430 $\Omega \pm 10\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 4-Load Branch MAF (Device Down) Topology [1] R1 - DATA: 18 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 30 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [2] R2- 51 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [3] R3- 4 Ω. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. It is optional to have R3 on the channel. It can be removed to reduce BOM cost. SPI0 2-Load Branch (Device Down) with G3 Isolation FET Topology [1] R1 (50 MHz) - DATA: 39 $\Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 22 $\Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_CLK. [2] R1 (80 MHz) - DATA: 18 $\Omega \pm 10\%$. To be placed within 7.62 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 10 $\Omega \pm 10\%$. To be placed within 7.62 mm from the branch on SPI0_CLK. SPI0 3-Load Branch (Device Down) with G3 EC Isolation FET of Flash Topology [1] R1 - DATA: 30 $\Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 22 $\Omega \pm 5\%$. To be placed within 7.62 mm from the branch on SPI0_CLK.

Signal Names	System Pull-up/Pull-down	Series Termination Resistor	Notes
			SPI0 3-Load Branch (Device Down) with G3 Isolation FET of Flash and TPM Topology [1] R1 (50 MHz) - $30\ \Omega \pm 5\%$. To be placed within 10.16mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R2 (50 MHz) - $51\ \Omega \pm 5\%$. To be placed within 10.16 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [3] R1 (80 MHz) - $10\ \Omega \pm 10\%$. To be placed within 10.16 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [4] R2 (80 MHz) - $430\ \Omega \pm 10\%$. To be placed within 10.16 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 Flash 80MHz 4-Load Branch (Device Down) with G3 Isolation FET Topology [1] R1- $10\ \Omega \pm 10\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R2- $430\ \Omega \pm 10\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 4-Load Branch (Device Down) with G3 Isolation FET Topology [1] R1 - $30\ \Omega \pm 10\%$. To be placed within 10.16 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R2 - $51\ \Omega \pm 10\%$. To be placed within 10.16 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 5-Load Branch (Device Down) with G3 Isolation FET Topology [1] R1 - DATA: $18\ \Omega \pm 5\%$. To be placed within 15.24 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: $30\ \Omega \pm 5\%$. To be placed within 15.24 mm from the branch on SPI0_CLK. [2] R2 -

Signal Names	System Pull-up/Pull-down	Series Termination Resistor	Notes
			51 $\Omega \pm 5\%$. To be placed within 15.24 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 2-Load Branch (Device Down) with G3 Wired-OR Topology [1] R1 - 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [2] R2 - 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [3] R3 - 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 3-Load Branch (Device Down) with G3 Wired-OR of Flash Topology [1] R1 - DATA: 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 10 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [2] R2 - DATA: 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 10 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [3] R3 - DATA: 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 10 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [4] R4 - DATA: 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 10 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. SPI0 3-Load Branch (Device Down) with G3 Wired-OR of Flash and TPM Topology [1] R1 - DATA: 10 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: 22 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [2] R2 - DATA: 10 $\Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.

Signal Names	System Pull-up/Pull-down	Series Termination Resistor	Notes
			CLK: $22\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [3] R3 - DATA: $10\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: $22\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [4] R4 - $51\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. SPI0 4-Load Branch (Device Down) with G3 Wired-OR Topology [1] R1 - DATA: $0\ \Omega$ placeholder. To be placed within 12.7 mm from the branch on SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. CLK: $15\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK. [2] R2 - $68\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [3] R3 - $15\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [4] R4 - $15\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3. [5] R5 - $68\ \Omega \pm 5\%$. To be placed within 12.7 mm from the branch on SPI0_CLK, SPI0_MISO, SPI0_MOSI, SPI0_IO2 and SPI0_IO3.

1.19 Touch Host Controller (THC) Interface Checklist

Table 27. Touch Host Controller Interface Checklist

Signal Names	Series Termination Resistor	Notes	✓
THC0_SPI_CLK	$18\ \Omega \pm 5\%$	To be placed 12.7 mm from the connector	
THC0_SPI_IO0	$18\ \Omega \pm 5\%$	To be placed 12.7 mm from the connector	
THC0_SPI_IO1	$18\ \Omega \pm 5\%$	To be placed 12.7 mm from the connector	
continued...			

Signal Names	Series Termination Resistor	Notes	✓
THC0_SPI_IO2	18 Ω $\pm$ 5%	To be placed 12.7 mm from the connector	
THC0_SPI_IO3	18 Ω $\pm$ 5%	To be placed 12.7 mm from the connector	
THC0_SPI_CS#		No series-resistor required.	
THC0_RST#		No series-resistor required.	
THC0_INT#		No series-resistor required.	
THC1_SPI_CLK	18 Ω $\pm$ 5%	To be placed 12.7 mm from the connector	
THC1_SPI_IO0	18 Ω $\pm$ 5%	To be placed 12.7 mm from the connector	
THC1_SPI_IO1	18 Ω $\pm$ 5%	To be placed 12.7 mm from the connector	
THC1_SPI_IO2	18 Ω $\pm$ 5%	To be placed 12.7 mm from the connector	
THC1_SPI_IO3	18 Ω $\pm$ 5%	To be placed 12.7 mm from the connector	
THC1_SPI_CS#		No series-resistor required.	
THC1_RST#		No series-resistor required.	
THC1_INT#		No series-resistor required.	
THC0_I2C_SDA THC1_I2C_SDA		The external pull-up resistor and current assist value depends on the speed supported and total bus capacitance. For more details, refer to the I2C / I3C chapter in the Nova Lake H Platform Design Guide (# 855310).	
THC0_I2C_SCL THC1_I2C_SCL		The external pull-up resistor and current assist value depends on the speed supported and total bus capacitance. For more details, refer to the I2C / I3C chapter in the Nova Lake H Platform Design Guide (# 855310).	

1.20 UART Checklist

Table 28. UART Checklist

Signal Names	System Pull-up/ Pull-down	Notes	✓
UART[2:0]_RXD UART[2:0]_TXD UART0_RTS# UART2_RTS# UART0_CTS# UART2_CTS#		If the interface is not used, the signals can be used as GPIO. If GPIO functionality is also not used, the signals can be left as no-connect. Bluetooth* devices are not supported on the Processor UART interfaces.	

1.21 GPIO Checklist

Care should be taken on GPIOs that default to a Native Function when they are used as GPIO function. These GPIOs should not be used if control/termination is required during Sx/Reset. These GPIO may be reset to the Native function on Sx/Reset (depending on GPIO PadRstCfg bit setting) with specific pin state, and may cause issue.

Refer to the Nova Lake GPIO Implementation Summary (# 852643) for details on GPIO multiplex and straps.

All unused GPIO pins that default to GPIO functionality do not need termination. For unused GPIO pins that default to a native function, termination on the pins depend on the termination requirement of the native function.

The reserved (RSVD) native functions linked to the GPIO pins must not be used for their original purpose. Instead, they can be reassigned for alternative native functions or configured as GPIO, as described in the Nova Lake H GPIO Implementation Summary Report.

1.22 Miscellaneous Signal Checklist

Refer to the Miscellaneous Signals chapter in the Nova Lake H Platform Design Guide (# 855310) for the connection guideline.

1.23 Intel® Converged Security Engine (Intel® CSE) Checklist

Table 29. ME-EC Interaction Checklist

Hardware Requirement	Expected Configuration	Verified (Y/N)	Comments
ME-EC Interaction for All Platforms Supporting Intel® CSE Corporate and Consumer Designs			
Intel® ME-EC: Intel® ME-EC connections according to latest revision of Intel® ME-EC Interaction Specification. Refer to the Intel® Converged Security and Manageability Engine and Embedded Controller Interaction Product Specification (# 626375) for more details on implementation.			
SLP_S3#	Optional for platforms with CM3 support. Required for platforms without CM3 support. If SLP_A# is routed from PCH to EC, then SLP_S3# can be optional from Intel® ME-EC perspective.		
SLP_A#	Required for platforms with CM3 support. SLP_A# should be routed from the PCH to the EC. SLP_A# asserted low informs the EC when Intel® CSE is in an CMOFF state. Depending on the platform, this pin may be used to control power to various devices that are part of the CSE sub-system in the platform.		
ACPRESENT	Required for systems with and without CM3. The ACPRESENT pin is used by EC for indicating		
<i>continued...</i>			

Hardware Requirement	Expected Configuration	Verified (Y/N)	Comments
	the current power source of the system to Intel® CSE. When asserted, it indicates that the platform is connected to an AC source.		

Table 30. Hardware and EC Firmware Requirements Checklist

Hardware Requirement	Expected Configuration	Verified (Y/N)	Comments
ME-EC Interaction for All Platforms Supporting Intel® Active Management Technology (Intel® AMT) Corporate Designs only			
Intel® ME-EC: Intel® ME-EC connections according to latest revision of Intel® ME-EC Interaction Specification. Refer to the Intel® Converged Security and Manageability Engine and Embedded Controller Interaction Product Specification (# 626375) for more details on implementation.			
SLP_A#	Required if platform supports CM3, otherwise optional.		
CL_CLK	Controller Link Clock: Bi-directional clock that connects to a Wireless LAN Device supporting Intel® Active Management Technology. For Controller Link design guidelines, refer to the Nova Lake H Platform Design Guide (# 855310).		
CL_DATA	Controller Link Data: Bi-directional data that connects to a Wireless LAN Device supporting Intel® Active Management Technology. For Controller Link design guidelines, refer to the Nova Lake H Platform Design Guide (# 855310).		
CL_RST#	Controller Link Reset: controller link reset pin that connects to a Wireless LAN Device supporting Intel® Active Management Technology. For Controller Link design guidelines, refer to the Nova Lake H Platform Design Guide (# 855310).		
SLP_LAN#	SLP_LAN# is asserted when power can be shut off to the PHY device(s). SLP_LAN# will always be de-asserted in S0 and anytime SLP_A# is de-asserted. <i>Note:</i> Power to LAN while it is powered must not be interrupted by either hardware or EC firmware on any of events (which include warm reset, power-cycle reset).		
TLS Confidentiality	Customer are expected to enable the strap.		

Table 31. Hardware and EC Firmware Requirements Checklist

Hardware Requirement	Expected Configuration	Verified (Y/N)	Comments
Platform Design Recommendation for All Platforms Supporting Intel® CSE Corporate and Consumer Designs			
Design Guide provides motherboard implementation recommendations for the platforms, based on the processor. Refer to the Nova Lake H Platform Design Guide (# 855310), or Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.			
SPI Write-Protect Pin	Must be pulled up to one of the following rails: A rail enabled by of SLP_LAN# or SLP_A#.		
SPI Flash Connectivity	SPI flash must be directly connected to the PCH for Intel® CSE. Intel recommends having a Flash socket for validation, but it is not required for production releases.		
SPI Flash Descriptor Security Override	GPP_B04 /BK0/SBK0/ISH_GP0 signal is used for Flash Descriptor Security Override (which is also known as Intel® CSE Debug Mode). This signal input should be tied to a jumper which connects to power through an external pull-up (1 KΩ ±5%) only. When this signal is asserted, the Flash Security gets overridden for ease of image programming. Refer to the Nova Lake-H/HX Processor External Design Specification, Volume 1 of 2, (# 844316) for more details.		
Power Sequence	Customers should ensure their platform follows the timing constraints in the "Platform Power Sequencing Specification" section in the Nova Lake H Platform Design Guide (# 855310).		
Intel® PCH XDP Debug port	Connect the PCH JTAG signals directly to PCH XDP debug port connector as described in the "Platform Debug and Test Hooks" chapter in the Nova Lake H Platform Design Guide (# 855310).		
Intel® APS Connectors	Customers are expected to follow the PDG and implement one of the two Intel® APS connectors as recommended in Intel® APS section in the Nova Lake H Platform Design Guide (# 855310).		

1.24 Intel® Integrated Sensor Solution (IISS) Checklist

This section provides schematics, PCB Layout and System Checklist for the Nova Lake IISS design to support in the design review and provide additional filter to catch design errors. Intel recommends that customer makes the best design according to the design guidelines provided runs over the following checklist and consults with the Intel CE, if there is any violation to the IISS Design Guidelines.

When to Use this Checklist?

- Schematics: to be performed after schematics is done before PCB Layout Design.
- PCB Layout: to be performed after chassis and PCB Layout are done before Gerber.
- System: to be performed after system is fully assembled and powered.

NOTE

Be noticed on your early designs. It is strongly recommended to reserve pads of series resistor and stuff with 0 Ohm resistors on ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1, ISH_I2C2 and ISH_GP[0..11,5A,6A,8A,9A,10A,11A]. Designers could separate sensor components onto ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 and ISH_I2C2. By doing this, it can keep flexibility of circuits, if an adjustment is needed on your PCB.

NOTE

For the Magnetometer, a special white paper was composed for supporting Magnetometer integration. Refer to the Mobile Computer Platform Magnetometer Design Integration Technical White Paper ([# 640580](#)).

1.24.1 Schematics Design Checklist

All labels may be prefixed with ISH_, if needed to differentiate them from other system schematic labels.

General Schematics Checklists

This section is used to record the actual sensors that the designer selected to use on the design. Based on the sensor used, the following tables should be filled by the designer with specific selected sensors data.

General	Verified (Y/N)	Comments
Did you read, understand and follow all design guidelines from your sensor manufacturers?		
Did you add layout instructions to the schematics for the layout designer to implement, such as keep-out zones for sensors, as mentioned in "System Debug" in Platform Design Guide?		
Specify the form factor type. (Clamshell, 2-in-1 Convertible/ 360-Clamshell, Tablet, Foldable, Detachable or others)		
Is ECI enabled at both ends of ISH and EC?		
Is IPCM enabled?		

Table 32. Accelerometer Sensor Details

Accelerometer Sensor	Verified (Y/N)	Comments
Sensor part number?		
Is it a combo-sensor?		
Where is the sensor been located? (Lid-side or keyboard-side)		
I2C/I3C bus the sensor is connected to? (ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 or ISH_I2C2)		
I2C/I3C address on the I2C/I3C bus?		
Is there an option to use alternative I2C/I3C Address? If so, what is the alternative I2C/I3C Address?		
Interrupt available?		
Interrupt polarity? (Active-High or Active-Low)		
Interrupt I/O buffer type? (Totem-Pole or Open-Drain)		
Is the same interrupt line shared with other component(s)? (ISH FW does not support it)		
VDD voltage? (1.8V or 3.3V)		
VDDIO voltage? (1.8V or 3.3V)		
Is a level-shifter placed between the sensor and ISH managed I2C/I3C bus?		
Is a level-shifter placed between the sensor and ISH managed GPIO?		

Table 33. Optional 2nd Accelerometer Sensor Details (for 2-in-1 Convertible/360-Clamshell/Foldable Design)

The 2nd Accelerometer Sensor	Verified (Y/N)	Comments
Sensor part number?		
Is it a combo-sensor?		
Where is the sensor been located? (Lid-side or keyboard-side)		
I2C/I3C bus the sensor is connected to? (ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 or ISH_I2C2)		
I2C/I3C address on the I2C/I3C bus?		
continued...		

The 2nd Accelerometer Sensor	Verified (Y/N)	Comments
Is there an option to use alternative I2C/I3C Address? If so, what is the alternative I2C/I3C Address?		
Interrupt available?		
Interrupt polarity? (Active-High or Active-Low)		
Interrupt I/O buffer type? (Totem-Pole or Open-Drain)		
Is the same interrupt line shared with other component(s)? (ISH FW does not support it)		
VDD voltage? (1.8V or 3.3V)		
VDDIO voltage? (1.8V or 3.3V)		
Is a level-shifter placed between the sensor and ISH managed I2C/I3C bus?		
Is a level-shifter placed between the sensor and ISH managed GPIO?		

Table 34. Gyroscope Sensor Details

Gyroscope Sensor	Verified (Y/N)	Comments
Sensor part number?		
Is it a combo-sensor?		
Where is the sensor been located? (Lid-side or keyboard-side)		
I2C/I3C bus the sensor is connected to? (ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 or ISH_I2C2)		
I2C/I3C address on the I2C/I3C bus?		
Is there an option to use alternative I2C/I3C Address? If so, what is the alternative I2C/I3C Address?		
VDD voltage? (1.8V or 3.3V)		
VDDIO voltage? (1.8V or 3.3V)		
Is a level-shifter placed between the sensor and ISH managed I2C/I3C bus?		

Table 35. Optional 2nd Gyroscope Sensor Details (for Foldable Design)

The 2nd Gyroscope Sensor	Verified (Y/N)	Comments
Sensor part number?		
Is it a combo-sensor?		
continued...		

The 2nd Gyroscope Sensor	Verified (Y/N)	Comments
Where is the sensor been located? (Lid-side or keyboard-side)		
I2C/I3C bus the sensor is connected to? (ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 or ISH_I2C2)		
I2C/I3C address on the I2C/I3C bus?		
Is there an option to use alternative I2C/I3C Address? If so, what is the alternative I2C/I3C Address?		
VDD voltage? (1.8V or 3.3V)		
VDDIO voltage? (1.8V or 3.3V)		
Is a level-shifter placed between the sensor and ISH managed I2C/I3C bus?		

Table 36. Magnetometer Sensor Details

Magnetometer Sensor	Verified (Y/N)	Comments
Sensor part number?		
Is it a combo-sensor?		
Where is the sensor been located? (Lid-side or keyboard-side)		
I2C/I3C bus the sensor is connected to? (ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 or ISH_I2C2)		
I2C/I3C address on the I2C/I3C bus?		
Is there an option to use alternative I2C/I3C Address? If so, what is the alternative I2C/I3C Address?		
VDD voltage? (1.8V or 3.3V)		
VDDIO voltage? (1.8V or 3.3V)		
Is a level-shifter placed between the sensor and ISH managed I2C/I3C bus?		

Table 37. Ambient Light Sensor (ALS) Details

Ambient Light Sensor (ALS)	Verified (Y/N)	Comments
Sensor part number?		
I2C/I3C bus the sensor is connected to? (ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 or ISH_I2C2)		
continued...		

Ambient Light Sensor (ALS)	Verified (Y/N)	Comments
I2C/I3C address on the I2C/I3C bus?		
Is there an option to use alternative I2C/I3C Address? If so, what is the alternative I2C/I3C Address?		
Interrupt available?		
Interrupt polarity? (Active-High or Active-Low)		
Interrupt I/O buffer type? (Totem-Pole or Open-Drain)		
Is the same interrupt line shared with other component(s)? (ISH FW does not support it)		
VDD voltage? (1.8V or 3.3V)		
VDDIO voltage? (1.8V or 3.3V)		
Is a level-shifter placed between the sensor and ISH managed I2C/I3C bus?		
Is a level-shifter placed between the sensor and ISH managed GPIO?		

Table 38. Pressure Sensor Details

Pressure Sensor	Verified (Y/N)	Comments
Sensor part number?		
I2C/I3C bus the sensor is connected to? (ISH_I3C0/ISH_I2C0, ISH_I3C1/ISH_I2C1 or ISH_I2C2)		
I2C/I3C address on the I2C/I3C bus?		
Is there an option to use alternative I2C/I3C Address? If so, what is the alternative I2C/I3C Address?		
Interrupt available?		
Interrupt polarity? (Active-High or Active-Low)		
Interrupt I/O buffer type? (Totem-Pole or Open-Drain)		
Is the same interrupt line shared with other component(s)? (ISH FW does not support it)		
VDD voltage? (1.8V or 3.3V)		
continued...		

Pressure Sensor	Verified (Y/N)	Comments
VDDIO voltage? (1.8V or 3.3V)		
Is a level-shifter placed between the sensor and ISH managed I2C/I3C bus?		
Is a level-shifter placed between the sensor and ISH managed GPIO?		

Table 39. Intel® Integrated Sensor Solution BOM Checklist

Intel Integrated Sensor Solution BOM	Verified (Y/N)	Comments
Are all the ISH managed sensors supported in IISS POR BOM? Specify them, if they are not in the BOM.		
Did you make sure your vendor provides support to you with IISS FDK, if the selected sensor(s) is/are not in IISS POR BOM?		
Did you make sure there is no conflict with I2C/I3C address on the same ISH I2C/I3C bus?		
Did you review the I2C/I3C design and make sure it conforms to the Platform Design Guide?		

Table 40. Sensors Debug Connector Checklist

Sensors Debug Connector	Verified (Y/N)	Comments
Is there any one of the two proposed debug connectors implemented in this design? Refer to "Sensors Debug Hooks" in the Platform Design Guide.		

Table 41. FW Debug Connector Checklist

FW Debug Connector	Verified (Y/N)	Comments
Was the XDP or the Primary CMC Debug Solution implemented? Refer to "Platform Debug and Test Hooks" in PDF for more details.		

Table 42. Power Rails Checklist

Power Rails	Verified (Y/N)	Comments
Did you check that all sensors connected to power rails with correct voltage level?		
Did you check that all sensors connected to ISH managed I2C/I3C interface(s) with correct voltage level? (Is a voltage level shifter needed?)		
Did you check that all sensors connected to ISH managed GPIO interface(s) with correct voltage level? (Is a voltage level shifter needed?)		

Table 43. Sensors I2C/I3C/Interrupt/GPIO Checklist

Sensors I2C/I3C/Interrupt/GPIO	Verified (Y/N)	Comments
Did you make sure there is no conflict with I2C/I3C address on the same ISH I2C/I3C bus?		
Did you correctly set the Address Selection pin on each of the sensors, to be accessible on its selected address?		
Did you check that all the sensors that are located on the same I2C/I3C bus operate at the selected frequency?		
Did you review the I2C/I3C design and make sure it conforms to the Platform Design Guide?		
Did you review the GPIO design and make sure it conforms to the Platform Design Guide?		

Table 44. Sensor SPI Schematics Checklist

Sensors SPI (ISH_SPI)	Verified (Y/N)	Comments
Did you follow the same requirement of GSPI?		

Table 45. Sensor UART Schematics Checklist

Sensors UART (ISH_UART[0..1])	Verified (Y/N)	Comments
Did you follow the same requirement of generic UART?		

1.25 Power Delivery Checklist

Table 46. Power Delivery Checklist

Signal Names	Voltage /Supply/PD Requirement	Value	Recommendation/ Remarks
VCC_CORE_SENSE	VCCCORE	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCC_CORE_VSS_SENSE	GND	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCCGT_SENSE	VCCGT	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCCGT_VSS_SENSE	GND	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCCSA_SENSE	VCCSA	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
continued...			

Signal Names	Voltage / Supply/PD Requirement	Value	Recommendation/ Remarks
VCCSA_VSS_SENSE	GND	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCCPRIM_VNNAON_VSS_SENSE	GND	10 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCCPRIM_IO_VSS_SENSE	GND	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCC_LP_ECORE_VSS_SENSE	GND	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCCPRIM_VNNAON_SENSE	VCCPRIM_VNNAON	10 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCCPRIM_IO_SENSE	VCCPRIM_IO	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty
continued...			

Signal Names	Voltage /Supply/PD Requirement	Value	Recommendation/ Remarks
			the feedback loop in open and the VR will be unstable without these "catch" resistors.
VCC_LP_ECORE_SENSE	VCC_LP_ECORE	100 ohm	Feedback "catch" resistors. Package sensing closes the feedback loop of the VR when a device is present in the CPU socket. When the CPU socket is empty the feedback loop in open and the VR will be unstable without these "catch" resistors.

1.26 Power Sequencing Checklist

Refer to the Nova Lake H Platform Design Guide (# 855310) for timing requirements.

Table 47. Interface Signals Names

Signals	Voltage /Supply Requirement	Recommendation	Remarks (Tick mark if implemented correct)
VCCRTC	1.2V	Connect RTC Coincell or Battery pack RTC of 3.3V to an ultra low bias current LDO to convert coincell voltage to 1.2V as required by processor.	
RTCRST# and SRTCRST#	VCCRTC	RTCRST# and SRTCRST# are connected to Processor with RC Delay to VCCRTC Supply. RC delay in the range of 18-25 ms. RC time delay for SRTCRST# should be the same as RC delay for RTCRST#. RTCRST# and SRTCRST# cannot be shorted together.	
VCCPRIM_VNNAON	0.77V	EN of VCCPRIM_VNNAON can be enabled by EC through its own logic. PG should be pulled up to 3.3V/1.8V according to platform or EC IO logic. Connect AND logic of PG of VCCPRIM_1P8 VR, VCCPRIM_3P3 VR as Enable signal of VCCPRIM_VNNAON VR. PG of VCCPRIM_VNNAON VR should be pulled up to VCCPRIM_1P8	
VCCPRIM_IO	1.25V	PG of VCCPRIM_VNNAON VR should be connected to EN of VCCPRIM_IO VR VR. PG of VCCPRIM_IO VR should be pulled up to 3.3V/1.8V according to platform or EC IO logic.	
VCCPRIM_1P8_PROC	1.8V	VCCPRIM_1P8_PROC VR should be enabled by PG of VCCPRIM_IO. PG of VCCPRIM_1P8_PROC VR should be pulled up to 3.3V/1.8V according to platform or EC IO logic.	
RSMRST#	1.25V	RSMRST# must be generated using PG of all PRIM rails logical AND with EN of VCCPRIM_VNNAON which is nothing but PRIM_VR_EN. IO voltage of RSMRST# is	
continued...			

Signals	Voltage / Supply Requirement	Recommendation	Remarks (Tick mark if implemented correct)
		1.25V. Hence it is suggested to use open drain buffer AND gate with 1.25V pull-up or a level translator to convert EC/platform IO logic to 1.25V. Add 2ms delay on rising edge of the signal. As this signal is input to Processor, Platform must ensure this signal doesn't glitch. Ensure RSMRST# can be driven low 400 ns before any of PRIM rails go out of spec in normal as well as surprise power down cases.	
VDD2	Memory tech dependant	This Power rail is enabled by SLP_S4#. PG of VDD2 VR should be pulled up to VCCPRIM_1P8. Follow JEDEC spec for both power-up and power down sequencing.	
VDDQ	Memory tech dependant	This Power rail is enabled by SLP_S4#. PG of VDDQ VR should be pulled up to VCCPRIM_1P8. Follow JEDEC spec for both power-up and power down sequencing.	
VCC_CORE, VCC_LP_ECOR E, VCCGT, VCCSA	Based on SVID	These Power rails are enabled by ALL_SYS_PWRGD with necessary delay through IMVP9.3.	
SLP_S5#		When asserted (Low) system is in S5. De-asserted in all other Sx states.	
SLP_S4#		When asserted (Low) system is in S4/S5. VDD2 must be enabled using this Signal. De-asserted in S3 and above states.	
SLP_S3#		When asserted (Low) system is in S3/S4/S5. De-asserted in S0. This SLP_S3# should be used as EN signal for VRs which should be ON in S0, S0ix states	
SLP_S0#		Asserted in S0ix state. De-asserted in all other Sx, S0 states.	
ALL_SYS_PWRGD		ALL_SYS_PWRGD is assumed to logically AND of SLP_S3#, RSMRST#, VDD2, VDDQ VR PG, S0 VR PGs signals for the major system power rails.	
PLT_PWROK	1.25V	VR_READY (Generated from IMVP9.3 VR) should be logical AND with ALL_SYS_PWRGD, SLP_S3# to generate PLT_PWROK. IO voltage of PLT_PWROK is 1.25V. Hence it is suggested to use open drain buffer AND gate with 1.25V pull-up or a level translator to convert EC/platform IO logic to 1.25V. As this signal is input to Processor, Platform must ensure this signal doesn't glitch. Ensure PLT_PWROK can be driven low 1us before VDD2 rail goes out of spec in normal as well as surprise power down cases. Also, PLT_PWROK should be driven low within 1us of SLP_S3# low. Hence it is better to use hardwire SLP_S3# in PLT_PWROK generation logic. VR_READY can be pulled up to 1.8V/3.3V/1.25V according to platform/EC logic.	
continued...			

Signals	Voltage /Supply Requirement	Recommendation	Remarks (Tick mark if implemented correct)
PLTRST#		Platform reset will be de-asserted from Processor and system will be out of reset. PLTRST# will be de-asserted (High) after PLT_PWROK High.	
PROC_C10_GATE#		Connect PROC_C10_GATE# to Test point for debug purpose. This signal will be asserted when Processor reaches C10 state.	
SLP_LAN#		Connect SLP_LAN# to LAN Device power to save power when not required. SLP_LAN# is used to indicate when power to the LAN PHY is needed. If Host Wake On LAN or Intel® ME Wake on LAN is enabled, the SLP_LAN# signal will remain high to keep power on to the LAN PHY.	
PWRBTN#		PWRBTN# need at least 16ms debouncing time. Hence PWRBTN# assertion time must be designed for minimum 18ms or more to be recognized. Signal driven from EC/Power Button to Processor indicating a system request to go into Sleep state OR if the system is already in the Sleep state, it will cause a wake event	
BATLOW#		If the BATLOW# signal is asserted, the Processor does not attempt to wake from an S4/S5 state, nor will it exit from pG3 state, even if the power button is pressed. This prevents the system from waking when the battery power is insufficient to wake the system. Wake events that occur while BATLOW# is asserted are latched by the Processor, and the system wakes after BATLOW# is de-asserted	
ACPRESENT		Used to indicate the presence of AC power or battery power. High - AC power is present. Low indicates Battery power only	
THERMTRIP#		Refer to the sideband chapter in the Nova Lake H Platform Design Guide (# 855310) for the connection guideline. Signal from the processor to indicate that a thermal overheating has occurred.	

1.27 Platform Debug and Test Hooks Checklist

Refer to the Chapter 4.7 in Open and Closed Chassis Debug Nova Lake Onwards User Guide (# 870129).

1.28 Power Management Checklist

Table 48. Power Management Checklist

Signals	System Pull-up / Pull-down	Schematics Notes	✓
PLTRST#		No requirements of pull up / pull down resistors for Glitch Free Implementation.	
THERMTRIP#	2.2 k Ω to 10 k Ω pull-up to VCCPRIM_1P8.	Pull-up required. A low value pull-up is OK since THERMTRIP# only goes low to shut down. In normal operation this signal is high.	
SLP_S3# SLP_S4# SLP_S5#		No requirements of pull up/pull down resistors for Glitch Free Implementation.	
SLP_S0#		No requirements of pull up/pull down resistors for Glitch Free Implementation.	
SLP_A#		No requirements of pull up/pull down resistors for Glitch Free Implementation. Can be left as NC when Intel® Active Management Technology (Intel® AMT) is not supported on the platform or there are no devices on the platform that is tied to Intel® AMT.	
SLP_LAN#		No requirements of pull up / pull down resistors for Glitch Free Implementation.	
PLT_PWROK	100k to 1M Ω pull-down to GND.	Pull-down required. If this signal is driven from an open drain source/ buffer the pull-up resistance should be in range 1/10th of pull down resistance to ensure the voltage is sufficient high enough to be seen as logic high by the platform. The pulldown is not needed if the source of this signal can guarantee this signal is in proper state for G3 through S0	
PWRBTN#		This signal has an internal pull-up resistor and has an internal 16 ms de-bounce on the input.	
<i>continued...</i>			

Signals	System Pull-up / Pull-down	Schematics Notes	✓
		<i>Note:</i> If the power button is powered in G3 state, when the Processor is unpowered, the platform must buffer the signal from the hard power button before it connects to the Processor PWRBTN# signal pin. The buffered version of PWRBTN# signal should be open drain and pulled up to the Processor VCCPRIM_1P8 rail. This is required to prevent leakage into the Processor in G3 states and when Processor is unpowered.	
SYS_RESET#	Connect this signal on Processor directly to the reset button and pull-up this signal to VCCPRIM_1P8 rail through a weak pull-up resistor (8.2 K Ω to 10K Ω). A 100-1 K Ω pull- down resistor isolated from SYS_RESET# by means of a normally open switch.	Pull-up required. SYS_RESET# is 1.8V signal and should be pulled-up to VCCPRIM_1P8 rail. Input to Processor cannot float. This pin forces an internal reset to the Processor after the signal is internally de- bounced. <i>Note:</i> SYS_RESET# is in the Primary Well but it will only affect the system when PLT_PWROK is high.	
RSMRST#	10k Ω to 100 k Ω pull- down to GND.	Pull-down required. RSMRST# is a RTC Well input and requires pull-down to reduce leakage from coin cell battery in G3. Input must not float in G3. <i>Note:</i> If this signal is driven from an open drain source/ buffer the pull-up resistance should be in range 1/10 th of pull-down resistance to ensure the voltage is sufficient high enough to be seen as logic high by the platform. Note that the pull-down is not needed if the source of this signal can guarantee this signal is in proper state for G3 through S0.	
GPP_V10/LANPHYPC		LANPHYPC should be connected to LAN_DISABLE_N on the PHY. Processor will drive LANPHYPC low to put the PHY into a low power state when functionality is not needed. Intel RVPs may have an empty pull-up resistor for testing purpose.	
continued...			

Signals	System Pull-up / Pull-down	Schematics Notes	✓
WAKE#	1 K Ω pull-up to VCCPRIM_1P8 rail.	The pull-up is required even if PCIe* interface is not used on the platform. Ensure that WAKE# signal Trise (Maximum) is <100 ns. Refer to the PCI Express* Mini- Card Electromechanical 2.0 Specification and PCI Express* Card Electromechanical 2.0 Specification for details.	
GPP_V00/BATLOW#	10 k Ω to 100 k Ω pull-up to VCCPRIM_1P8 rail.	Pull-up required even if not implemented.	
GPP_V01/ACPRESENT	10 k Ω pull-up to VCCPRIM_1P8 rail.	Pull-up required. Point-to-point connection between Processor and EC. To indicate that system is powered by AC. This is a required signal for platforms with or without M3 support.	
PROC_C10_GATE#		No requirements of pull up/pull down resistors for Glitch Free Implementation.	
DRAM_RESET#	470 Ω pull-up place holder to VDD2	Pull-up place holder is required.	
PMCALERT#	10K Ω Pull-up to VCCPRIM_1P8 Rail	Pull-up is required regardless of whether Integrated USB-C is used or not.	
VDD2PWRGOOD_OUT		Connect VDD2PWRGOOD_OUT to VDD2PWRGOOD_IN through a 0 ohm Resistor	
VDD2PWRGOOD_IN		Connect VDD2PWRGOOD_OUT to VDD2PWRGOOD_IN through a 0 ohm Resistor	